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Assignment 2

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Question 1

We are given the general element of $so(5)$

$$X = x_1 L_{12} + x_2 L_{13} + x_3 L_{14} + x_4 L_{23} + x_5 L_{24} + x_6 L_{23} + x_7 L_{25} + x_8 L_{34} + x_9 L_{35} + x_{10} L_{45} \quad (1)$$

The adjoint action is given by the commutator of the generators and the general element namely:

$$\text{ad}_X (L_{ij}) = [X, L_{ij}] \quad (2)$$

Given that $L_{ij} = E_{ji} - E_{ij}$ and

We can look at the sub algebra h with the general element

$$Z = z_1 L_{13} + z_2 L_{24} \quad (3)$$

$$\begin{aligned} [L_{ij}, L_{kl}] &= [E_{ji} - E_{ij}, E_{lk} - E_{kl}] \\ &= [E_{ji}, E_{lk}] - [E_{ji}, E_{kl}] - [E_{ij}, E_{lk}] + [E_{ij}, E_{kl}] \\ &= \delta_{il} E_{jk} - \delta_{jk} E_{li} - \delta_{ik} E_{jl} + \delta_{jl} E_{ki} - \delta_{jl} E_{ik} + \delta_{ik} E_{lj} + \delta_{jk} E_{il} - \delta_{il} E_{kj} \\ &= \delta_{il} (E_{jk} - E_{kj}) - \delta_{ik} (E_{jl} - E_{lj}) - \delta_{jk} (E_{li} - E_{il}) + \delta_{jl} (E_{ki} - E_{ik}) \\ [L_{ij}, L_{kl}] &= \delta_{il} L_{kj} - \delta_{ik} L_{lj} - \delta_{jk} L_{il} + \delta_{jl} L_{ik}. \end{aligned}$$

And we have the adjoint given by

$$\begin{aligned}
\text{ad}_Z L_{ij} &= [Z, L_{ij}] \\
&= z_1 \underbrace{[L_{13}, L_{ij}]}_{\Lambda_1} + z_2 \underbrace{[L_{24}, L_{ij}]}_{\Lambda_2} \\
\text{ad}_Z L_{12} &= z_1 [L_{13}, L_{12}] + z_2 [L_{24}, L_{12}] \\
&= z_2 L_{41} - z_1 L_{23} \\
\text{ad}_Z L_{13} &= z_1 [L_{13}, L_{13}] + z_2 [L_{24}, L_{13}] \\
&= 0 \\
\text{ad}_Z L_{14} &= z_1 [L_{13}, L_{14}] + z_2 [L_{24}, L_{14}] \\
&= z_2 L_{21} - z_1 L_{43} \\
\text{ad}_Z L_{15} &= z_1 [L_{13}, L_{15}] + z_2 [L_{24}, L_{15}] \\
&= -z_1 L_{53} \\
\text{ad}_Z L_{23} &= z_1 [L_{13}, L_{23}] + z_2 [L_{24}, L_{23}] \\
&= z_1 L_{12} - z_2 L_{34} \\
\text{ad}_Z L_{24} &= z_1 [L_{13}, L_{24}] + z_2 [L_{24}, L_{24}] \\
&= 0 \\
\text{ad}_Z L_{25} &= z_1 [L_{13}, L_{25}] + z_2 [L_{24}, L_{25}] \\
&= -z_2 L_{54} \\
\text{ad}_Z L_{34} &= z_1 [L_{13}, L_{34}] + z_2 [L_{24}, L_{34}] \\
&= z_2 L_{23} - z_1 L_{14} \\
\text{ad}_Z L_{35} &= z_1 [L_{13}, L_{35}] + z_2 [L_{24}, L_{35}] \\
&= -z_1 L_{15} \\
\text{ad}_Z L_{45} &= z_1 [L_{13}, L_{45}] + z_2 [L_{24}, L_{45}] \\
&= -z_2 L_{25}
\end{aligned}$$

$$\begin{aligned}
[Z, L_{12}] &= z_2 L_{14} - z_1 L_{23}, \\
[Z, L_{13}] &= 0, \\
[Z, L_{14}] &= -z_2 L_{12} + z_1 L_{34}, \\
[Z, L_{15}] &= z_1 L_{35}, \\
[Z, L_{23}] &= z_1 L_{12} - z_2 L_{34}, \\
[Z, L_{24}] &= 0, \\
[Z, L_{25}] &= z_2 L_{45}, \\
[Z, L_{34}] &= -z_1 L_{14} + z_2 L_{23}, \\
[Z, L_{35}] &= -z_1 L_{15}, \\
[Z, L_{45}] &= -z_2 L_{25}.
\end{aligned}$$

Thus if we use the basis

$$(L_{12}, L_{13}, L_{14}, L_{15}, L_{23}, L_{24}, L_{25}, L_{34}, L_{35}, L_{45}) \quad (4)$$

We get

$$\text{ad}_Z = \begin{bmatrix} 0 & 0 & -z_2 & 0 & z_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ z_2 & 0 & 0 & 0 & 0 & 0 & 0 & -z_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -z_1 & 0 \\ -z_1 & 0 & 0 & 0 & 0 & 0 & 0 & z_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -z_2 \\ 0 & 0 & z_1 & 0 & -z_2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & z_2 & 0 & 0 & 0 \end{bmatrix} \quad (5)$$

Part 2

To find the eigenvalues and eigenvectors I used mathematica (see file).

For the subspace spanned by $\{L_{25}, L_{45}\}$:

$$\begin{aligned} \text{Eigenvalues} &= \lambda_{\pm} = \pm iz_2 \\ \text{Eigenvectors} &= U_{\pm} = L_{25} \pm iL_{45}. \end{aligned}$$

For the subspace spanned by $\{L_{15}, L_{35}\}$

$$\begin{aligned} \text{Eigenvalues } \lambda_{\pm} &= \pm iz_1 \\ \text{Eigenvectors} &= I_{\pm} = L_{15} \pm iL_{35} \end{aligned}$$

Finally for the subspace spanned by $\{L_{12}, L_{14}, L_{34}, L_{23}\}$:

We have the adjoint given by

$$\text{ad}_Z = \begin{bmatrix} 0 & -z_2 & 0 & z_1 \\ z_2 & 0 & -z_1 & 0 \\ 0 & z_1 & 0 & -z_2 \\ -z_1 & 0 & z_2 & 0 \end{bmatrix} \quad (6)$$

Which gives the eigenvalues

$$\lambda = \pm i (z_1 + z_2), \quad \lambda = \pm i (z_1 - z_2) \quad (7)$$

we get the eigenvectors given by

$$V_{\pm}^{-} = L_{14} + L_{23} \mp i (L_{12} + L_{34}) \text{ corresponding to } \lambda = \pm i (z_1 - z_2) \quad (8)$$

and

$$V_{\pm}^{+} = L_{14} + L_{23} \pm i (L_{34} - L_{12}) \text{ corresponding to } \lambda = \pm i (z_1 + z_2) \quad (9)$$

$$v_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = L_{24}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = L_{13}, \quad v_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -i \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix},$$

$$v_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ i \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_5 = \begin{pmatrix} -1 \\ 0 \\ -i \\ 0 \\ i \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_6 = \begin{pmatrix} 1 \\ 0 \\ i \\ 0 \\ i \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

$$v_7 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -i \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad v_8 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ i \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad v_9 = \begin{pmatrix} 1 \\ 0 \\ -i \\ 0 \\ -i \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

$$v_{10} = \begin{pmatrix} -1 \\ 0 \\ i \\ 0 \\ -i \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

$$\begin{aligned}
v_1 &= L_{24}, & v_2 &= L_{13}, & v_3 &= -iL_{15} + L_{35}, & v_4 &= iL_{15} + L_{35} \\
v_5 &= -L_{12} - iL_{14} + iL_{23} + L_{34}, & v_6 &= L_{12} + iL_{14} + iL_{23} + L_{34}, & v_7 &= -iL_{25} + L_{45} \\
v_8 &= iL_{25} + L_{45} & v_9 &= L_{12} - iL_{14} - iL_{23} + L_{34}, & v_{10} &= -L_{12} + iL_{14} - iL_{23} + L_{34}.
\end{aligned}$$

Thus the normalised eigenvectors are

$$\begin{aligned}
V_{\pm}^+ &= c_{V^+} [L_{23} - L_{14} \pm i(L_{34} - L_{12})] \\
V_{\pm}^- &= c_{V^-} [L_{14} + L_{23} \mp i(L_{12} + L_{34})] \\
I_{\pm} &= c_I [L_{15} \pm iL_{35}] \\
U_{\pm} &= c_U [L_{25} \pm iL_{45}].
\end{aligned}$$

A general element in the cartan subalgebra is given by

$$Z = (z_1, z_2) \Xi^{-1} \begin{bmatrix} H_1 \\ H_2 \end{bmatrix}.$$

$$\begin{aligned}
[Z, I_{\pm}] &= \lambda_{\pm} I_{\pm} \\
\begin{bmatrix} z_1 & z_2 \end{bmatrix} \Xi^{-1} [H, I_{\pm}] &= \lambda_{\pm} I_{\pm}.
\end{aligned}$$

For our cartan eigenvalues we had $\lambda_{\pm}^I = \pm iz_1$.

$$\begin{aligned}
[z_1, z_2] \Xi^{-1} \begin{bmatrix} H & I_{\pm} \end{bmatrix} &= \pm iz_1 I_{\pm} \\
\begin{bmatrix} z_1 & z_2 \end{bmatrix} \Xi^{-1} \begin{bmatrix} H, & I_{\pm} \end{bmatrix} &= \pm i \begin{bmatrix} z_1 & z_2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} I_{\pm} \\
[H, I_{\pm}] &= \underbrace{\pm i \Xi \begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\vec{\lambda}} I_{\pm} = \pm i \Xi \begin{bmatrix} 1 \\ 0 \end{bmatrix} (L_{15} \pm iL_{35}).
\end{aligned}$$

For the subalgebra spanned by $\{L_{25}, L_{45}\}$ with eigenvalues $\lambda_{\pm}^U = \pm iz_2$. We can do the same process to get

$$[H, U_{\pm}] = \underbrace{\pm i \Xi \begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\vec{\lambda}} U_{\pm} = \pm i \Xi \begin{bmatrix} 0 \\ 1 \end{bmatrix} (L_{25} \pm iL_{45}).$$

For the subspace spanned by $\{L_{12}, L_{14}, L_{34}, L_{23}\}$ with eigenvalues $\lambda = \pm i(z_1 + z_2)$

and $\lambda = \pm i(z_1 - z_2)$.

For the first pair:

$$\begin{aligned} \begin{bmatrix} z_1 & z_2 \end{bmatrix} \Xi^{-1} \begin{bmatrix} H & I_{\pm} \end{bmatrix} &= (\pm i(z_1 + z_2)) V_{\pm} \\ \begin{bmatrix} z_1 & z_2 \end{bmatrix} \Xi^{-1} \begin{bmatrix} H & V_{\pm} \end{bmatrix} &= \pm i \begin{bmatrix} z_1 & z_2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} V_{\pm} \\ [H, V_{\pm}] &= \underbrace{\pm i \Xi \begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\vec{\lambda}} V_{\pm} = \pm i \Xi \begin{bmatrix} 1 \\ -1 \end{bmatrix} (L_{12} - L_{14} \pm i(L_{34} - L_{12})) \end{aligned}$$

For the second pair:

$$\begin{aligned} \begin{bmatrix} z_1 & z_2 \end{bmatrix} \Xi^{-1} \begin{bmatrix} H & I_{\pm} \end{bmatrix} &= (\pm i(z_1 - z_2)) V_{\pm} \\ \begin{bmatrix} z_1 & z_2 \end{bmatrix} \Xi^{-1} \begin{bmatrix} H & V_{\pm} \end{bmatrix} &= \pm i \begin{bmatrix} z_1 & z_2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} V_{\pm} \\ [H, V_{\pm}] &= \underbrace{\pm i \Xi \begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{\vec{\lambda}} V_{\pm} = \pm i \Xi \begin{bmatrix} 1 \\ -1 \end{bmatrix} (L_{14} - L_{23} \mp i(L_{12} + L_{34})) \end{aligned}$$

Thus our roots are given by

$$\vec{\lambda}_{U_{\pm}} = \pm \Xi \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \vec{\lambda}_{I_{\pm}} = \pm \Xi \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \vec{\lambda}_{V_{\pm}^+} = \pm \Xi \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \vec{\lambda}_{V_{\pm}^-} = \pm \Xi \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Part 3 (a)

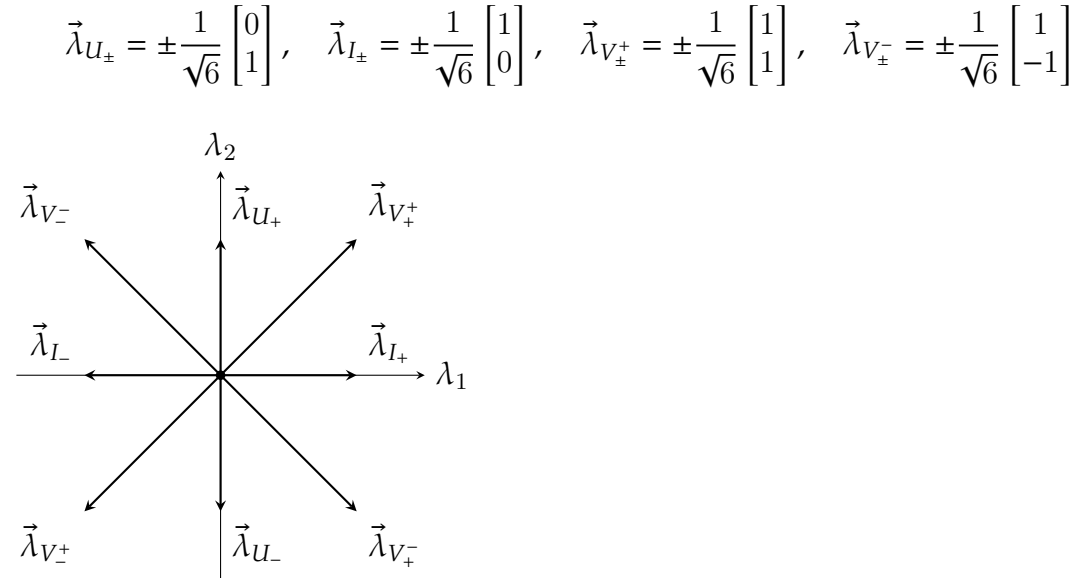
We are given that $K(X, X) = -6x^2$.

Thus $K(Z, Z) = -6(z_1^2 + z_2^2) = -6\mathbf{1}_{2 \times 2}$. So we have that

$$\Xi = -\frac{i}{\sqrt{6}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{10}$$

Thus all the roots will be scaled by $\frac{-i}{\sqrt{6}}$.

$$\begin{aligned}
[H, U_{\pm}] &= \underbrace{\pm \Xi \begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\vec{\lambda}} i U_{\pm} = \pm i \frac{1}{\sqrt{6}} \begin{bmatrix} 0 \\ 1 \end{bmatrix} (L_{25} \pm i L_{45}) \\
[H, I_{\pm}] &= \underbrace{\pm \Xi \begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\vec{\lambda}} i I_{\pm} = \pm i \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} (L_{15} \pm i L_{35}) \\
[H, V_{\pm}] &= \underbrace{\pm \Xi \begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\vec{\lambda}} i V_{\pm} = \pm i \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} (L_{12} - L_{14} \pm i (L_{34} - L_{12})) \\
[H, V_{\pm}] &= \underbrace{\pm \Xi \begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{\vec{\lambda}} i V_{\pm} = \pm i \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} (L_{14} - L_{23} \mp i (L_{12} + L_{34}))
\end{aligned}$$



Part b

We have

$$K(H, H) = \sum_s \vec{\lambda}_s \vec{\lambda}_s^T.$$

If the roots are α_i then we have the following where we notice for the negative roots we get an even power on a minus sign so we can just double the positive roots.

$$\begin{aligned}
K &= \sum_{i=1}^4 \left(\frac{\alpha_i}{\sqrt{6}} \right) \left(\frac{\alpha_i}{\sqrt{6}} \right)^T + \left(\frac{-\alpha_i}{\sqrt{6}} \right) \left(\frac{-\alpha_i}{\sqrt{6}} \right)^T \\
&= \frac{2}{6} \sum_{i=1}^4 (\alpha_i) (\alpha_i)^T \\
&= \frac{1}{3} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} \right) \\
&= \frac{1}{3} \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right) \\
\sum_s \lambda_s \lambda_s^T &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.
\end{aligned}$$

Now for

$$S = \sum_s \vec{\lambda}_s \cdot \vec{\lambda}_s \quad (11)$$

$$S = \frac{1}{6} \sum_s^8 \alpha_s \cdot \alpha_s = 2 \cdot \frac{1}{6} (1 + 1 + 2 + 2) = 2.$$

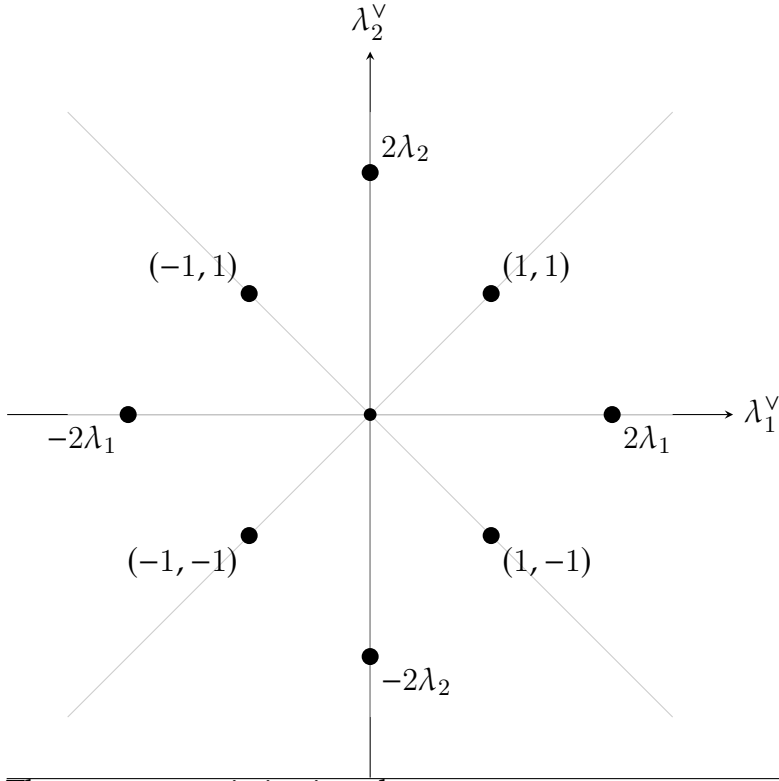
Part c

The co-roots are given by

$$\tilde{\lambda}^\wedge = \frac{2\lambda}{\langle \lambda, \lambda \rangle} \quad (12)$$

$$\langle \lambda_1, \lambda_1 \rangle = \langle \lambda_2, \lambda_2 \rangle = 1 \quad \langle \lambda_3, \lambda_3 \rangle = \langle \lambda_4, \lambda_4 \rangle = 2 \quad (13)$$

$$\begin{aligned}
\tilde{\lambda}_1^\wedge &= \pm \sqrt{6} \cdot 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \pm 2\sqrt{6} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
\tilde{\lambda}_2^\wedge &= \pm \sqrt{6} \cdot 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \pm 2\sqrt{6} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
\tilde{\lambda}_3^\wedge &= \pm \sqrt{6} \frac{2}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \pm \sqrt{6} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
\tilde{\lambda}_4^\wedge &= \pm \sqrt{6} \frac{2}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \pm \sqrt{6} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.
\end{aligned}$$



The cartan matrix is given by

$$A_{(i,j)} = 2 \frac{\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_j, \alpha_j \rangle} \quad (14)$$

For the simple roots we can pick a long root $\alpha_1 = [1 \ -1]$ and a short root $\alpha_2 = [0 \ 1]$

$$\begin{aligned} \alpha_1 \cdot \alpha_1 &= 2, & \alpha_2 \cdot \alpha_2 &= 1 & \alpha_1 \cdot \alpha_2 &= -1 \\ A_{11} &= \frac{2}{2} = 2 \\ A_{12} &= -2 \\ A_{21} &= -1 \\ A_{22} &= 2 \\ A_{(i,j)} &= \begin{bmatrix} 2 & -1 \\ -2 & 2 \end{bmatrix}. \end{aligned}$$

The positive roots are $+\alpha = \{[1 \ 0], [0 \ 1], [1 \ 1], [1 \ -1]\}$ These can be wrtten as

$$+\alpha = \{\alpha_1 + \alpha_2, \alpha_2, \alpha_1 + 2\alpha_2, \alpha_1\} \quad (15)$$

The simple co-roots are given by

$$\alpha_1^\wedge = [1, -1] \quad \alpha_2^\wedge = 2 [0 \ 1] \quad (16)$$

And the other co-roots are given by

$$(\alpha_1 + \alpha_2)^\wedge = \frac{2(\alpha_1 + \alpha_2)}{\langle (\alpha_1 + \alpha_2), (\alpha_1 + \alpha_2) \rangle} = 2 \frac{\begin{bmatrix} 1 & 0 \end{bmatrix}}{1} = 2 \begin{bmatrix} 1 & 0 \end{bmatrix} = 2(\alpha_1 + \alpha_2)$$

$$(\alpha_1 + 2\alpha_2)^\wedge = \frac{2(\alpha_1 + 2\alpha_2)}{\langle (\alpha_1 + 2\alpha_2), (\alpha_1 + 2\alpha_2) \rangle} = 2 \cdot \frac{\begin{bmatrix} 1 & 1 \end{bmatrix}}{2} = \begin{bmatrix} 1 & 1 \end{bmatrix} = \alpha_1 + \alpha_2$$

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The co-roots in terms of the simple co-roots are

$$\begin{aligned}\alpha_1^\vee &= 1 \cdot \alpha_1^\vee + 0 \cdot \alpha_2^\vee, \\ \alpha_2^\vee &= 0 \cdot \alpha_1^\vee + 1 \cdot \alpha_2^\vee, \\ (\alpha_1 + \alpha_2)^\vee &= 2 \cdot \alpha_1^\vee + 1 \cdot \alpha_2^\vee, \\ (\alpha_1 + 2\alpha_2)^\vee &= 1 \cdot \alpha_1^\vee + 1 \cdot \alpha_2^\vee.\end{aligned}$$

part e

The fundamental weights ω_j is given by

$$\tilde{\alpha}_i^\wedge \cdot \omega_j = 2 \frac{\langle \tilde{\alpha}_i, \tilde{w}_j \rangle}{\langle \tilde{\alpha}_i, \tilde{\alpha}_i \rangle} = \delta_{ij} \quad (17)$$

Take the co-roots $\lambda_1^\wedge = \begin{bmatrix} 1 & -1 \end{bmatrix}^T$, $\alpha_2^\wedge = \begin{bmatrix} 0 & 1 \end{bmatrix}^T$ Then if $\omega_1 = (x_1, y_1)$ we have

$$\begin{aligned}x_1 - y_1 &= \frac{1}{\sqrt{6}} \\ 2y_1 &= 0\end{aligned}$$

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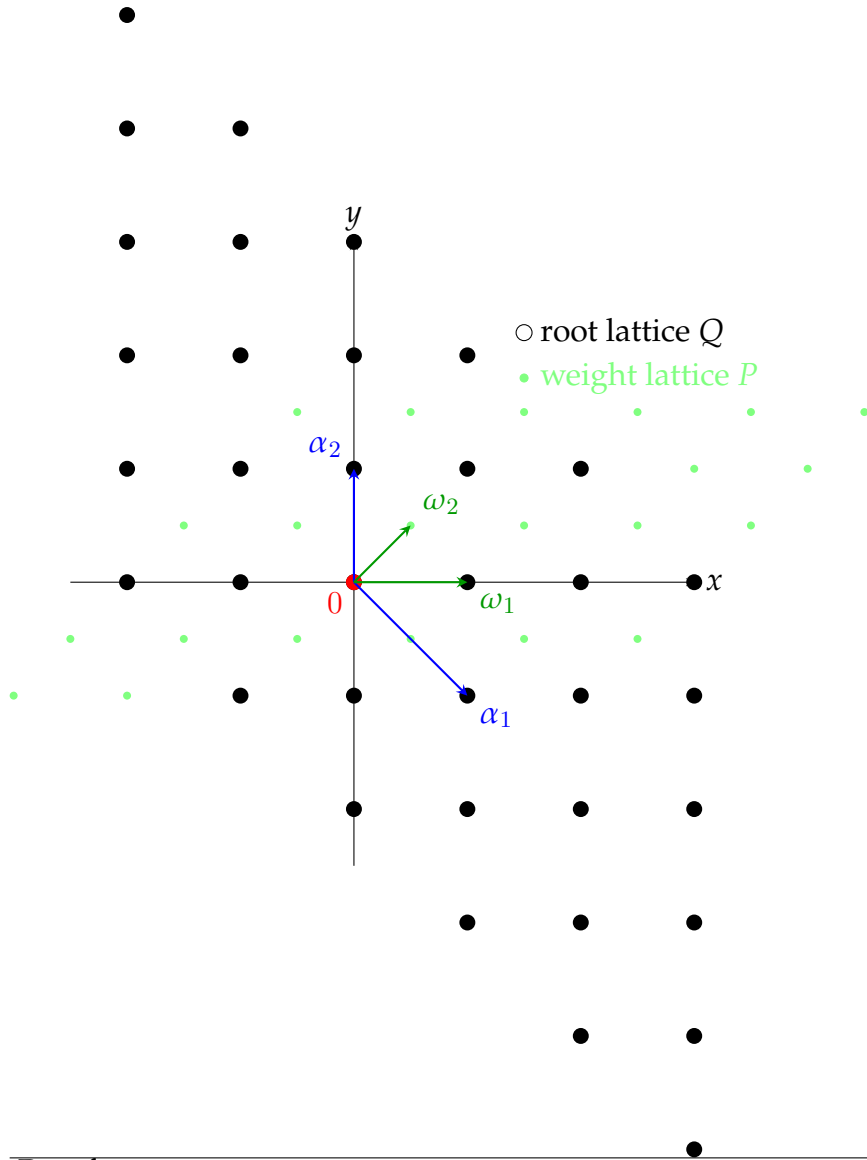
So $\omega_1 = \frac{1}{\sqrt{6}} (1, 0)$

For $\omega_2 = (x_2, y_2)$

$$\begin{aligned}x_2 - y_2 &= 0 \\ 2y_2 &= \frac{1}{\sqrt{6}}.\end{aligned}$$

So $\omega_2 = \frac{1}{\sqrt{6}} (\frac{1}{2}, \frac{1}{2})$

Which can be seen below



Part f

We need to impose the constraints

$$K(E_{\vec{\lambda}}, E_{-\vec{\lambda}}) = \frac{2}{|\vec{\lambda}|^2}, \quad E_{\lambda} = E_{\vec{\lambda}}^+.$$

We have that

$$K(L_{ij}, L_{ij}) = -6.$$

The short roots $S = \{\alpha_2, \alpha_1 + \alpha_2\}$ have $|\lambda|^2 = \frac{1}{6}$ so we have

$$K_S(E_{\vec{\lambda}}, E_{-\vec{\lambda}}) = 12.$$

The long roots $L = \{\alpha_1, \alpha_1 + 2\alpha_2\}$ have $|\lambda|^2 = \frac{1}{3}$ so we have

$$K_L(E_{\vec{\lambda}}, E_{-\vec{\lambda}}) = -6.$$

So for the short roots we can find the normalisation coefficient for one of them and it will hold for all of them (likewise for the long ones).

Look at α_2 corresponds to $U_{\pm} = L_{25} \pm iL_{45}$

$$K_S(E_{U_+}, E_{U_-}) = 6$$

$$K_S(U_+, U_-) = K_S(L_{25} + iL_{45}, L_{25} - iL_{45}) = K_S(L_{25}, L_{25}) + K_S(L_{45}, L_{45})$$

$$K_S(U_+, U_-) = -6 - 6 = -12 \quad \text{and} \quad E_{\alpha_2} = c_S U_+ \quad E_{-\alpha_2} = -c_S^* U_-$$

$$K_S(E_{\alpha_2}, E_{-\alpha_2}) = -|c_S|^2 K_S(U_+, U_-) = -|c_S|^2 \cdot -12 = 12$$

$$|c_S|^2 = 1 \text{ we can look at } c_S = i.$$

For the long roots we know α_1 corresponds to $\{L_{12}, L_{14}, L_{34}, L_{23}\}$.

$$V_{\pm} = L_{14} + L_{23} \mp i(L_{12} + L_{34})$$

$$K_L(E_{V_+}, E_{V_-}) = 6$$

$$K_L(V_+, V_-) = K_L(L_{14} + L_{23} - i(L_{12} + L_{34}), L_{14} + L_{23} + i(L_{12} + L_{34}))$$

$$K_L(V_+, V_-) = K_L(L_{14}, L_{14}) + K_L(L_{23}, L_{23}) + K_L(L_{12}, L_{12}) + K_L(L_{34}, L_{34})$$

$$K_L(V_+, V_-) = 4(-6) = -24 \quad K_L(E_{\alpha_1}, E_{-\alpha_2}) = -|c_L|^2 K_L(V_+, V_-) = -|c_L|^2 (-24)$$

$$|c_L|^2 (24) = 6 \quad |c_L|^2 = \frac{1}{4}$$

$$\text{so pick } c_L = \frac{i}{2}.$$

Thus we can choose the following normalisations for the root vectors

$$E_{\alpha_2} = i(-L_{25} + iL_{45}) = L_{45} - iL_{25}$$

$$E_{\alpha_1+\alpha_2} = i(L_{15} - iL_{35}) = L_{35} + iL_{15}$$

$$E_{\alpha_1} = \frac{i}{2}(L_{14} + L_{23} - i(L_{12} + L_{34})) = \frac{1}{2}(L_{12} + L_{34} + i(L_{14} + L_{23}))$$

$$E_{\alpha_1+2\alpha_2} = \frac{i}{2}(L_{23} - L_{14} + i(-L_{34} + L_{12})) = \frac{1}{2}(L_{34} - L_{12} + i(L_{23} - L_{14}))$$

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Using the hermitian conjugates for the negative roots.

$$E_{-\alpha_2} = L_{45} + iL_{25}$$

$$E_{-(\alpha_1+\alpha_2)} = L_{35} - iL_{15}$$

$$E_{-\alpha_1} = \frac{1}{2}(L_{12} + L_{34} - i(L_{14} + L_{23}))$$

$$E_{-(\alpha_1+2\alpha_2)} = \frac{1}{2}(-L_{34} - L_{12} - i(L_{23} - L_{14}))$$

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part g

We choose the (normalised) root vectors

$$E_{\alpha_2} = L_{45} - iL_{25}, \quad E_{-\alpha_2} = L_{45} + iL_{25},$$

and

$$\begin{aligned} E_{\alpha_1} &= \frac{1}{2}(L_{12} + L_{34} + i(L_{14} + L_{23})), \\ E_{-\alpha_1} &= \frac{1}{2}(L_{12} + L_{34} - i(L_{14} + L_{23})), \\ E_{\alpha_1+\alpha_2} &= L_{35} + iL_{15}, \quad E_{-(\alpha_1+\alpha_2)} = L_{35} - iL_{15}, \\ E_{\alpha_1+2\alpha_2} &= \frac{1}{2}(-L_{12} - iL_{14} + iL_{23} + L_{34}), \\ E_{-(\alpha_1+2\alpha_2)} &= \frac{1}{2}(-L_{12} + iL_{14} - iL_{23} + L_{34}), \end{aligned}$$

with Cartan generators

$$H_1 = L_{13}, \quad H_2 = L_{24}.$$

We use

$$[L_{ij}, L_{kl}] = \delta_{il}L_{kj} - \delta_{ik}L_{lj} - \delta_{jk}L_{il} + \delta_{jl}L_{ik}. \quad (18)$$

We take the (normalised) root vectors

$$E_{\alpha_2} = L_{45} - iL_{25}, \quad E_{-\alpha_2} = L_{45} + iL_{25}, \quad (19)$$

$$E_{\alpha_1} = \frac{1}{2}(L_{12} + L_{34} + i(L_{14} + L_{23})), \quad E_{-\alpha_1} = \frac{1}{2}(L_{12} + L_{34} - i(L_{14} + L_{23})), \quad (20)$$

$$E_{\alpha_1+\alpha_2} = L_{35} + iL_{15}, \quad E_{-(\alpha_1+\alpha_2)} = L_{35} - iL_{15}, \quad (21)$$

$$E_{\alpha_1+2\alpha_2} = \frac{1}{2}(L_{34} - L_{12} + i(L_{23} - L_{14})), \quad E_{-(\alpha_1+2\alpha_2)} = \frac{1}{2}(L_{34} - L_{12} - i(L_{23} - L_{14})). \quad (22)$$

$$[E_{\alpha_2}, E_{-\alpha_2}]$$

$$\begin{aligned} [E_{\alpha_2}, E_{-\alpha_2}] &= [L_{45} - iL_{25}, L_{45} + iL_{25}] \\ &= [L_{45}, L_{45}] + i[L_{45}, L_{25}] - i[L_{25}, L_{45}] - i^2[L_{25}, L_{25}] \\ &= 0 + i(-L_{24}) - i(L_{24}) - 0 \\ &= -iL_{24} - iL_{24} \\ &= -2iL_{24}. \end{aligned}$$

Thus

$$[E_{\alpha_2}, E_{-\alpha_2}] = -2iH_2,$$

with $H_2 = L_{24}$.

$$[E_{\alpha_2}, E_{-\alpha_1}]$$

$$E_{-\alpha_1} = \frac{1}{2}(L_{12} + L_{34} - i(L_{14} + L_{23})).$$

Then

$$\begin{aligned} [E_{\alpha_2}, E_{-\alpha_1}] &= \frac{1}{2}[L_{45} - iL_{25}, L_{12} + L_{34} - i(L_{14} + L_{23})] \\ &= \frac{1}{2}([L_{45}, L_{12}] + [L_{45}, L_{34}] - i[L_{45}, L_{14}] - i[L_{45}, L_{23}] \\ &\quad - i[L_{25}, L_{12}] - i[L_{25}, L_{34}] + i^2[L_{25}, L_{14}] + i^2[L_{25}, L_{23}]). \end{aligned}$$

Using the L_{ij} -commutators above:

$$\begin{aligned} [L_{45}, L_{12}] &= 0, \quad [L_{45}, L_{34}] = L_{35}, \quad [L_{45}, L_{14}] = L_{15}, \quad [L_{45}, L_{23}] = 0, \\ [L_{25}, L_{12}] &= L_{15}, \quad [L_{25}, L_{34}] = 0, \quad [L_{25}, L_{14}] = 0, \quad [L_{25}, L_{23}] = -L_{35}, \end{aligned}$$

we get

$$\begin{aligned} [E_{\alpha_2}, E_{-\alpha_1}] &= \frac{1}{2}(0 + L_{35} - iL_{15} - 0 - iL_{15} - 0 + 0 + (-1)(-L_{35})) \\ &= \frac{1}{2}(2L_{35} - 2iL_{15}) \\ &= L_{35} - iL_{15}. \end{aligned}$$

Since $E_{-(\alpha_1+\alpha_2)} = L_{35} - iL_{15}$, we obtain

$$[E_{\alpha_2}, E_{-\alpha_1}] = E_{-(\alpha_1+\alpha_2)}.$$

$$[E_{\alpha_2}, E_{\alpha_1}]$$

$$E_{\alpha_1} = \frac{1}{2}(L_{12} + L_{34} + i(L_{14} + L_{23})).$$

Then

$$\begin{aligned} [E_{\alpha_2}, E_{\alpha_1}] &= \frac{1}{2}[L_{45} - iL_{25}, L_{12} + L_{34} + i(L_{14} + L_{23})] \\ &= \frac{1}{2}([L_{45}, L_{12}] + [L_{45}, L_{34}] + i[L_{45}, L_{14}] + i[L_{45}, L_{23}] \\ &\quad - i[L_{25}, L_{12}] - i[L_{25}, L_{34}] - i^2[L_{25}, L_{14}] - i^2[L_{25}, L_{23}]). \end{aligned}$$

Using the same L_{ij} -commutators,

$$\begin{aligned} [E_{\alpha_2}, E_{\alpha_1}] &= \frac{1}{2}(0 + L_{35} + iL_{15} + 0 - iL_{15} - 0 - (-1) \cdot 0 - (-1)(-L_{35})) \\ &= \frac{1}{2}(L_{35} + iL_{15} - iL_{15} - L_{35}) \\ &= 0. \end{aligned}$$

Hence

$$[E_{\alpha_2}, E_{\alpha_1}] = 0.$$

$$[E_{\alpha_2}, E_{\alpha_1+\alpha_2}]$$

$$E_{\alpha_1+\alpha_2} = L_{35} + iL_{15}.$$

Then

$$\begin{aligned} [E_{\alpha_2}, E_{\alpha_1+\alpha_2}] &= [L_{45} - iL_{25}, L_{35} + iL_{15}] \\ &= [L_{45}, L_{35}] + i[L_{45}, L_{15}] - i[L_{25}, L_{35}] - i^2[L_{25}, L_{15}]. \end{aligned}$$

Using

$$[L_{45}, L_{35}] = -L_{34}, \quad [L_{45}, L_{15}] = -L_{14}, \quad [L_{25}, L_{35}] = L_{23}, \quad [L_{25}, L_{15}] = -L_{12},$$

we get

$$\begin{aligned} [E_{\alpha_2}, E_{\alpha_1+\alpha_2}] &= (-L_{34}) + i(-L_{14}) - i(L_{23}) - (-1)(-L_{12}) \\ &= -L_{34} - iL_{14} - iL_{23} - L_{12} \\ &= -(L_{12} + L_{34} + i(L_{14} + L_{23})) \\ &= -2E_{\alpha_1}. \end{aligned}$$

Thus

$$[E_{\alpha_2}, E_{\alpha_1+\alpha_2}] = -2E_{\alpha_1}.$$

$$[E_{\alpha_2}, E_{-(\alpha_1+\alpha_2)}]$$

$$E_{-(\alpha_1+\alpha_2)} = L_{35} - iL_{15}.$$

Then

$$\begin{aligned} [E_{\alpha_2}, E_{-(\alpha_1+\alpha_2)}] &= [L_{45} - iL_{25}, L_{35} - iL_{15}] \\ &= [L_{45}, L_{35}] - i[L_{45}, L_{15}] - i[L_{25}, L_{35}] + (-i)(-i)[L_{25}, L_{15}] \\ &= -L_{34} - i(-L_{14}) - i(L_{23}) + (-1)(-L_{12}) \\ &= -L_{34} + iL_{14} - iL_{23} + L_{12}. \end{aligned}$$

Now

$$E_{\alpha_1+2\alpha_2} = \frac{1}{2}(L_{34} - L_{12} + i(L_{23} - L_{14})),$$

so

$$\begin{aligned} -2E_{\alpha_1+2\alpha_2} &= -(L_{34} - L_{12} + i(L_{23} - L_{14})) \\ &= -L_{34} + L_{12} + i(-L_{23} + L_{14}) \\ &= -L_{34} + L_{12} + iL_{14} - iL_{23}, \end{aligned}$$

which matches the expression above. Hence

$$[E_{\alpha_2}, E_{-(\alpha_1+\alpha_2)}] = -2E_{\alpha_1+2\alpha_2}.$$

$$[E_{\alpha_2}, E_{\alpha_1+2\alpha_2}]$$

$$E_{\alpha_1+2\alpha_2} = \frac{1}{2}(L_{34} - L_{12} + i(L_{23} - L_{14})).$$

Then

$$\begin{aligned} [E_{\alpha_2}, E_{\alpha_1+2\alpha_2}] &= \frac{1}{2}[L_{45} - iL_{25}, L_{34} - L_{12} + i(L_{23} - L_{14})] \\ &= \frac{1}{2}([L_{45}, L_{34}] - [L_{45}, L_{12}] + i[L_{45}, L_{23}] - i[L_{45}, L_{14}] \\ &\quad - i[L_{25}, L_{34}] + i[L_{25}, L_{12}] + [L_{25}, L_{23}] - [L_{25}, L_{14}]). \end{aligned}$$

Using the known commutators,

$$[L_{45}, L_{34}] = L_{35}, \quad [L_{45}, L_{12}] = 0, \quad [L_{45}, L_{23}] = 0, \quad [L_{45}, L_{14}] = L_{15},$$

$$[L_{25}, L_{34}] = 0, \quad [L_{25}, L_{12}] = L_{15}, \quad [L_{25}, L_{23}] = -L_{35}, \quad [L_{25}, L_{14}] = 0,$$

we get

$$\begin{aligned} [E_{\alpha_2}, E_{\alpha_1+2\alpha_2}] &= \frac{1}{2}(L_{35} - 0 + i \cdot 0 - iL_{15} - i \cdot 0 + iL_{15} - L_{35} - 0) \\ &= \frac{1}{2}(L_{35} - iL_{15} + iL_{15} - L_{35}) \\ &= 0. \end{aligned}$$

Thus

$$[E_{\alpha_2}, E_{\alpha_1+2\alpha_2}] = 0.$$

$$[E_{\alpha_2}, E_{-(\alpha_1+2\alpha_2)}]$$

$$E_{-(\alpha_1+2\alpha_2)} = \frac{1}{2}(L_{34} - L_{12} - i(L_{23} - L_{14})) = \frac{1}{2}(L_{34} - L_{12} - iL_{23} + iL_{14}).$$

Then

$$\begin{aligned} [E_{\alpha_2}, E_{-(\alpha_1+2\alpha_2)}] &= \frac{1}{2}[L_{45} - iL_{25}, L_{34} - L_{12} - iL_{23} + iL_{14}] \\ &= \frac{1}{2}([L_{45}, L_{34}] - [L_{45}, L_{12}] - i[L_{45}, L_{23}] + i[L_{45}, L_{14}] \\ &\quad - i[L_{25}, L_{34}] + i[L_{25}, L_{12}] + [L_{25}, L_{23}] - [L_{25}, L_{14}]). \end{aligned}$$

Using the same commutators as above,

$$\begin{aligned} [E_{\alpha_2}, E_{-(\alpha_1+2\alpha_2)}] &= \frac{1}{2}(L_{35} - 0 - i \cdot 0 + iL_{15} - i \cdot 0 + iL_{15} - (-L_{35}) - 0) \\ &= \frac{1}{2}(2L_{35} + 2iL_{15}) \\ &= L_{35} + iL_{15}. \end{aligned}$$

Since $E_{\alpha_1+\alpha_2} = L_{35} + iL_{15}$, this gives

$$[E_{\alpha_2}, E_{-(\alpha_1+2\alpha_2)}] = E_{\alpha_1+\alpha_2}.$$

The non-vanishing commutators of E_{α_2} with the Weyl elements are

$$\begin{aligned} [E_{\alpha_2}, E_{-\alpha_2}] &= -2i H_2, \\ [E_{\alpha_2}, E_{-\alpha_1}] &= E_{-(\alpha_1+\alpha_2)}, \\ [E_{\alpha_2}, E_{\alpha_1+\alpha_2}] &= -2 E_{\alpha_1}, \\ [E_{\alpha_2}, E_{-(\alpha_1+\alpha_2)}] &= -2 E_{\alpha_1+2\alpha_2}, \\ [E_{\alpha_2}, E_{-(\alpha_1+2\alpha_2)}] &= E_{\alpha_1+\alpha_2}, \end{aligned}$$

while all other commutators with E_{α_2} vanish:

$$\begin{aligned} [E_{\alpha_2}, E_{\alpha_2}] &= 0, \\ [E_{\alpha_2}, E_{\alpha_1}] &= 0, \\ [E_{\alpha_2}, E_{\alpha_1+2\alpha_2}] &= 0. \end{aligned}$$

In particular,

$$[E_{\alpha_2}, E_{-\alpha_2}] = \alpha_2^\vee \cdot \vec{H} = -2i H_2,$$

in agreement with the standard relation $[E_\lambda, E_{-\lambda}] = \lambda^\vee \cdot \vec{H}$.

Part 4 a

$$A = \begin{bmatrix} 2 & -1 \\ -2 & 2 \end{bmatrix}.$$

$$\vec{\alpha}_1 = \vec{\lambda}_4 = i \Xi \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}, \quad \vec{\alpha}_2 = \vec{\lambda}_2 = i \Xi \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

$$\Xi = -i \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix},$$

$$\begin{bmatrix} H_1 \\ H_2 \end{bmatrix} = \Xi \begin{bmatrix} L_{13} \\ L_{24} \end{bmatrix} = -i \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} L_{13} \\ L_{24} \end{bmatrix} = \begin{bmatrix} -iL_{13} + iL_{24} \\ -2iL_{24} \end{bmatrix}.$$

$$\lambda_4 = \alpha_1, \quad \lambda_2 = \alpha_2, \quad \lambda_1 = \alpha_1 + \alpha_2, \quad \lambda_3 = \alpha_1 + 2\alpha_2.$$

Using

$$\vec{\alpha}_1 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}, \quad \vec{\alpha}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix},$$

we obtain the positive roots

$$\vec{\lambda}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \vec{\lambda}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \vec{\lambda}_3 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad \vec{\lambda}_4 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}.$$

Part 4 b

$$-6 \Xi \Xi^T = 6 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} = 12 \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

$$g^{-1} = 12 \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix},$$

$$g = \frac{1}{12} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Define

$$\tilde{\lambda} = g \lambda,$$

so that

$$|\lambda|^2 = \tilde{\lambda} \cdot \lambda.$$

$$\lambda_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \lambda_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \lambda_3 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad \lambda_4 = \begin{bmatrix} 2 \\ -2 \end{bmatrix},$$

we find

$$\tilde{\lambda}_1 = g \lambda_1 = \frac{1}{12} \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \tilde{\lambda}_2 = g \lambda_2 = \frac{1}{12} \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

$$\tilde{\lambda}_3 = g\lambda_3 = \frac{1}{6} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \tilde{\lambda}_4 = g\lambda_4 = \frac{1}{6} \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The corresponding lengths are

$$|\lambda_1|^2 = \frac{1}{6}, \quad |\lambda_2|^2 = \frac{1}{6}, \quad |\lambda_3|^2 = \frac{1}{3}, \quad |\lambda_4|^2 = \frac{1}{3}.$$

The coroots are given by

$$\tilde{\lambda}^\vee = 2 \frac{\tilde{\lambda}}{|\tilde{\lambda}|^2}.$$

$$\tilde{\lambda}_1^\vee = (2, 1), \quad \tilde{\lambda}_2^\vee = (0, 1), \quad \tilde{\lambda}_3^\vee = (1, 1), \quad \tilde{\lambda}_4^\vee = (1, 0).$$

Since $\alpha_1 = \lambda_4$ and $\alpha_2 = \lambda_2$, the simple coroots are

$$\tilde{\alpha}_1^\vee = \tilde{\lambda}_4^\vee = (1, 0), \quad \tilde{\alpha}_2^\vee = \tilde{\lambda}_2^\vee = (0, 1).$$

Part 4 c

The fundamental weights ω_j are defined by

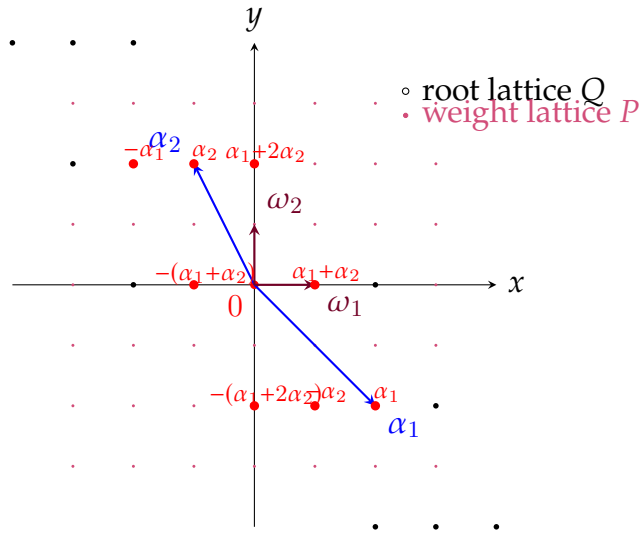
$$\tilde{\alpha}_i^\vee \cdot \omega_j = \delta_{ij},$$

where the dot is the normal dot product.

$$\omega_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \omega_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Thus the weight lattice is the $\mathbb{Z}$ -span of $\{\omega_1, \omega_2\}$, and the root lattice is the $\mathbb{Z}$ -span of the simple roots $\{\alpha_1, \alpha_2\}$, with

$$\alpha_1 = (2, -2), \quad \alpha_2 = (-1, 2).$$



Question 2

$\frac{\alpha_1}{\alpha_2} = \sqrt{3}$. Angle between is 150° .

The number of roots $\Lambda = \dim g - r$. Here we have $\dim G_2 = 14$ and we have 12 roots. So the rank is $r = 2$.

We have 6 long roots and 6 short roots with the ratio $\sqrt{3}$. So

$$6a^2 + 6(\sqrt{3}a)^2 = 2$$

$$24a^2 = 2 \quad a^2 = \frac{1}{12}.$$

Thus the short roots have length $\ell_s = \frac{1}{\sqrt{12}} = \frac{1}{2\sqrt{3}}$ and the long roots have length $\ell_l = \sqrt{3} \cdot \ell_s = \frac{1}{2}$.

From this we know that $\alpha_2 = \frac{1}{2\sqrt{3}} \begin{bmatrix} 0 & 1 \end{bmatrix}$, and $\alpha_1 = \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix}$

From this we know that

$$\alpha_2 = \frac{1}{2\sqrt{3}} \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad \alpha_1 = \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix}.$$

Now we can look at the positive roots.

$$\begin{aligned}\alpha_1 + 3\alpha_2 &= \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix} + 3 \cdot \frac{1}{2\sqrt{3}} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 & 2\sqrt{3} \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 1 & \sqrt{3} \end{bmatrix},\end{aligned}$$

$$\begin{aligned}\alpha_1 + 2\alpha_2 &= \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix} + 2 \cdot \frac{1}{2\sqrt{3}} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 & \frac{4}{\sqrt{3}} \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 1 & \frac{\sqrt{3}}{3} \end{bmatrix},\end{aligned}$$

$$\begin{aligned}2\alpha_1 + 3\alpha_2 &= 2 \cdot \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix} + 3 \cdot \frac{1}{2\sqrt{3}} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix} + \frac{\sqrt{3}}{2} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 0 \end{bmatrix},\end{aligned}$$

$$\begin{aligned}\alpha_1 + \alpha_2 &= \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix} + \frac{1}{2\sqrt{3}} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 & \frac{2}{\sqrt{3}} \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 1 & -\frac{\sqrt{3}}{3} \end{bmatrix}.\end{aligned}$$

Part 2

The coroots are given by

$$\tilde{\lambda}^\wedge = \frac{2\lambda}{\langle \lambda, \lambda \rangle} \tag{23}$$

$$\tilde{\alpha}_1^\wedge = \frac{2 \cdot \frac{1}{4}}{\frac{1}{4}} \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix}^T = 2 \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix}^T =$$

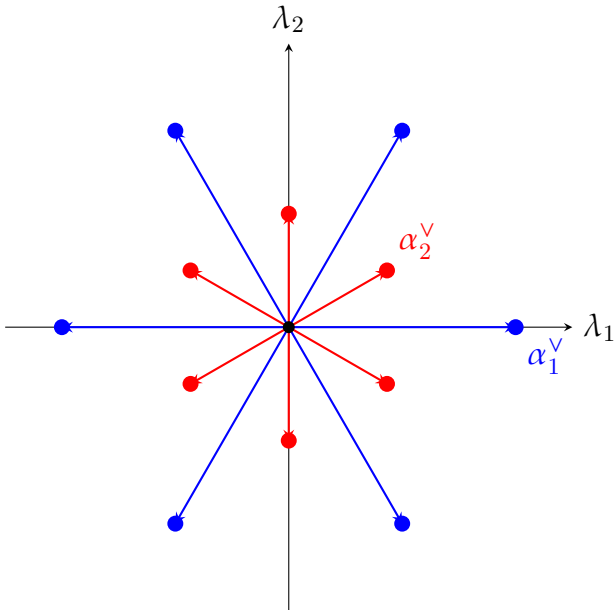
$$\tilde{\alpha}_2^\wedge = \frac{2 \cdot \frac{1}{2\sqrt{3}}}{\frac{1}{12}} \begin{bmatrix} 1 & 0 \end{bmatrix}^T = 4\sqrt{3} \begin{bmatrix} 0 & 1 \end{bmatrix}^T$$

$$(\alpha_1 + 3\alpha_2)^\wedge = 2 \frac{\frac{1}{4} \begin{bmatrix} 1 & \sqrt{3} \end{bmatrix}^T}{\frac{1}{4}} = 2 \begin{bmatrix} 1 & \sqrt{3} \end{bmatrix}^T = \alpha_1^\wedge + \alpha_2^\wedge$$

$$(\alpha_1 + 2\alpha_2)^\wedge = 2 \frac{\frac{1}{4} \begin{bmatrix} 1 & \frac{\sqrt{3}}{3} \end{bmatrix}^T}{\frac{1}{12}} = 6 \begin{bmatrix} 1 & \frac{\sqrt{3}}{3} \end{bmatrix}^T = 3\alpha_1^\wedge + 2\alpha_2^\wedge$$

$$(2\alpha_1 + 3\alpha_2)^\wedge = 2 \frac{\frac{1}{2} \begin{bmatrix} 1 & 0 \end{bmatrix}^T}{\frac{1}{4}} = 4 \begin{bmatrix} 1 & 0 \end{bmatrix}^T = 2\alpha_1^\wedge + \alpha_2^\wedge$$

$$(\alpha_1 + \alpha_2)^\wedge = 2 \frac{\frac{1}{4} \begin{bmatrix} 1 & -\frac{\sqrt{3}}{3} \end{bmatrix}^T}{\frac{1}{12}} = 6 \begin{bmatrix} 1 & -\frac{\sqrt{3}}{3} \end{bmatrix}^T = 3\alpha_1^\wedge + \alpha_2^\wedge.$$



The elements of the cartan matrix are given by

$$A_{(i,j)} = \tilde{\alpha}_i^\wedge \cdot \tilde{\alpha}_j = 2 \frac{\langle \tilde{\alpha}_i, \tilde{\alpha}_j \rangle}{\langle \tilde{\alpha}_i, \tilde{\alpha}_i \rangle}.$$

$$A_{11} = \frac{1}{2} (1 + 3) = 2$$

$$A_{12} = \frac{1}{\sqrt{3}} (-\sqrt{3}) = -1$$

$$A_{21} = \sqrt{3} \cdot -\sqrt{3} = -3$$

$$A_{22} = 2$$

$$A = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}.$$

Part 3

$$[\vec{H}, E_{\vec{\alpha}}] = \vec{\alpha} E_{\vec{\alpha}} \quad (24)$$

We have

$$[\vec{H}, E_{\vec{\alpha}_1}] = [H_1, E_{\vec{\alpha}_1}] + [H_2, E_{\vec{\alpha}_1}].$$

where the cartan subalgebra is given by $\vec{H} = \begin{bmatrix} H_1 \\ H_2 \end{bmatrix}$ and $H_1 = 2 \begin{bmatrix} 1 & -\sqrt{3} \end{bmatrix}$, $H_2 = 4\sqrt{3} \begin{bmatrix} 0 & 1 \end{bmatrix}$

$$[H_1, E_{\vec{\alpha}_1}] = \frac{1}{4} E_{\vec{\alpha}_1}, \quad [H_2, E_{\vec{\alpha}_1}] = -\frac{\sqrt{3}}{4} E_{\vec{\alpha}_1}$$

$$[H_1, E_{\vec{\alpha}_2}] = 0, \quad [H_2, E_{\vec{\alpha}_2}] = \frac{1}{2\sqrt{3}} E_{\vec{\alpha}_2}$$

$$[H_1, E_{\vec{\alpha}_1+3\vec{\alpha}_2}] = \frac{1}{4} E_{\vec{\alpha}_1+3\vec{\alpha}_2}, \quad [H_2, E_{\vec{\alpha}_1+3\vec{\alpha}_2}] = \frac{\sqrt{3}}{4} E_{\vec{\alpha}_1+3\vec{\alpha}_2}$$

$$[H_1, E_{\vec{\alpha}_1+2\vec{\alpha}_2}] = \frac{1}{4} E_{\vec{\alpha}_1+2\vec{\alpha}_2}, \quad [H_2, E_{\vec{\alpha}_1+2\vec{\alpha}_2}] = \frac{1}{4\sqrt{3}} E_{\vec{\alpha}_1+2\vec{\alpha}_2}$$

$$[H_1, E_{2\vec{\alpha}_1+3\vec{\alpha}_2}] = \frac{1}{2} E_{2\vec{\alpha}_1+3\vec{\alpha}_2}, \quad [H_2, E_{2\vec{\alpha}_1+3\vec{\alpha}_2}] = 0$$

$$[H_1, E_{\vec{\alpha}_1+\vec{\alpha}_2}] = \frac{1}{4} E_{\vec{\alpha}_1+\vec{\alpha}_2}, \quad [H_2, E_{\vec{\alpha}_1+\vec{\alpha}_2}] = -\frac{1}{4\sqrt{3}} E_{\vec{\alpha}_1}.$$

Part 4

$$[E_{\vec{\alpha}}, E_{-\vec{\alpha}}] = \tilde{\alpha}^\wedge \cdot \vec{H} \quad (25)$$

$$\begin{aligned}
[E_{\vec{\alpha}_1}, E_{-\vec{\alpha}_1}] &= 2H_1 - 2\sqrt{3}H_2 \\
[E_{\vec{\alpha}_2}, E_{-\vec{\alpha}_2}] &= +4\sqrt{3}H_2 \\
[E_{\vec{\alpha}_1+3\vec{\alpha}_2}, E_{-(\vec{\alpha}_1+3\vec{\alpha}_2)}] &= 2H_1 + 2\sqrt{3}H_2 \\
[E_{\vec{\alpha}_1+2\vec{\alpha}_2}, E_{-(\vec{\alpha}_1+2\vec{\alpha}_2)}] &= 6H_1 + 2\sqrt{3}H_2 \\
[E_{2\vec{\alpha}_1+3\vec{\alpha}_2}, E_{-(2\vec{\alpha}_1+3\vec{\alpha}_2)}] &= 4H_1 \\
[E_{\vec{\alpha}_1+\vec{\alpha}_2}, E_{-(\vec{\alpha}_1+\vec{\alpha}_2)}] &= 6H_1 - 2\sqrt{3}H_2.
\end{aligned}$$

Part 5

I will pick the first chain where $\vec{\alpha} = \vec{\alpha}_2$ and $\vec{\beta} = \vec{\alpha}_1$.

We can use this for the non-collinear commutators which are given by

$$[E_{\vec{\alpha}}, E_{\vec{\beta}}] = N_{\vec{\alpha}+\vec{\beta}} E_{\vec{\alpha}+\vec{\beta}}.$$

We can construct the chain using $\vec{\beta} + k\vec{\alpha}$ for $k \in \mathbb{Z}$ and the maximum and minimum can be identified using the root diagram. $m = -k_{\min} = 0$ and $n = k_{\max} = 3$. So $k \in \{0, 1, 2, 3\}$

We have

$$\begin{aligned}
|n_{\vec{\alpha}, \vec{\beta}}|^2 &= \frac{1}{2} n(m+1) \langle \vec{\alpha}, \vec{\alpha} \rangle \\
|n_{\vec{\alpha}, \vec{\beta}+k\vec{\alpha}}|^2 &= \frac{1}{2} (n-k)(m+k+1) \langle \vec{\alpha}, \vec{\alpha} \rangle \\
|n_{\vec{\alpha}_2, \vec{\alpha}_1+k\vec{\alpha}_2}|^2 &= \frac{1}{2} (3-k)(k+1) \cdot \frac{1}{12}.
\end{aligned}$$

We get the values

$$\begin{aligned}
k=0 \quad |n_{\vec{\alpha}_2, \vec{\alpha}_1}|^2 &= \frac{1}{2} 3(1) \cdot \frac{1}{12} = \frac{1}{8} \\
k=1 \quad |n_{\vec{\alpha}_2, \vec{\alpha}_1+\vec{\alpha}_2}|^2 &= \frac{1}{2} (3-1)(1+1) \cdot \frac{1}{12} = \frac{1}{6} \\
k=2 \quad |n_{\vec{\alpha}_2, \vec{\alpha}_1+2\vec{\alpha}_2}|^2 &= \frac{1}{2} (3-2)(2+1) \cdot \frac{1}{12} = \frac{1}{8} \\
k=3 \quad |n_{\vec{\alpha}_2, \vec{\alpha}_1+3\vec{\alpha}_2}|^2 &= \frac{1}{2} (3-3)(3+1) \cdot \frac{1}{12} = 0
\end{aligned}$$

We can use the discrete symmetries (note this is a different γ)

$$\begin{aligned}
n_{\vec{\beta}, \vec{\alpha}}^* &= -n_{\vec{\alpha}, \vec{\beta}} \quad n_{\vec{\beta}, \vec{\alpha}}^* = -n_{-\vec{\alpha}, -\vec{\beta}} \\
n_{\vec{\beta}, \vec{\alpha}} &= n_{\vec{\beta}, \vec{\gamma}} = n_{\vec{\gamma}, \vec{\alpha}} \iff \vec{\alpha} + \vec{\beta} + \vec{\gamma} = \vec{0}.
\end{aligned}$$

Thus we have

$$\begin{aligned}
|n_{\vec{\alpha}_2, \vec{\alpha}_1}| &= |n_{-\vec{\alpha}_2, -\vec{\alpha}_1}| = \frac{\sqrt{2}}{4} & n_{\vec{\alpha}_1, \vec{\alpha}_2} &= -\frac{\sqrt{2}}{4} \\
|n_{\vec{\alpha}_2, \vec{\alpha}_1 + \vec{\alpha}_2}| &= |n_{-\vec{\alpha}_2, -\vec{\alpha}_1 - \vec{\alpha}_2}| = \frac{1}{\sqrt{6}} & n_{\vec{\alpha}_1 + \vec{\alpha}_2, \alpha_2} &= -\frac{1}{\sqrt{6}} \\
|n_{\vec{\alpha}_2, \vec{\alpha}_1 + 2\vec{\alpha}_2}| &= |n_{-\vec{\alpha}_2, -\vec{\alpha}_1 - 2\vec{\alpha}_2}| = \frac{\sqrt{2}}{4} & n_{\vec{\alpha}_1 + 2\vec{\alpha}_2, \alpha_2} &= -\frac{\sqrt{2}}{4} \\
|n_{\vec{\alpha}_2, \vec{\alpha}_1 + 3\vec{\alpha}_2}| &= |n_{-\vec{\alpha}_2, -\vec{\alpha}_1 - 3\vec{\alpha}_2}| = 0 & n_{\vec{\alpha}_1 + 3\vec{\alpha}_2, \vec{\alpha}_2} &= 0
\end{aligned}$$

We can use

$$E_{\vec{\alpha}} = \sqrt{\frac{2\eta_{\vec{\alpha}}}{\langle \vec{\alpha}, \vec{\alpha} \rangle}} e_{\vec{\alpha}} \quad (26)$$

Thus from now on I will work with e 's which is equivalent up to a factor
We have the relation

$$[e_{\vec{\alpha}}, e_{\vec{\beta}}] = n_{\vec{\alpha}, \vec{\beta}} e_{\vec{\alpha} + \vec{\beta}}.$$

Thus we get the commutators

$$\begin{aligned}
[e_{\vec{\alpha}_1}, e_{\vec{\alpha}_1}] &= [e_{\vec{\alpha}_2}, e_{\vec{\alpha}_2}] = 0 \\
[e_{\vec{\alpha}_1}, e_{\vec{\alpha}_2}] &= -\frac{\sqrt{2}}{4} e_{\vec{\alpha}_1 + \vec{\alpha}_2} & [e_{-\vec{\alpha}_1}, e_{-\vec{\alpha}_2}] &= -\frac{\sqrt{2}}{4} e_{-\vec{\alpha}_1 - \vec{\alpha}_2} \\
[e_{\vec{\alpha}_2}, e_{\vec{\alpha}_1 + \vec{\alpha}_2}] &= -\frac{1}{\sqrt{6}} e_{\vec{\alpha}_1 + 2\vec{\alpha}_2} & [e_{-\vec{\alpha}_2}, e_{-\vec{\alpha}_1 - \vec{\alpha}_2}] &= -\frac{1}{\sqrt{6}} e_{-\vec{\alpha}_1 - 2\vec{\alpha}_2} \\
[e_{\vec{\alpha}_2}, e_{\vec{\alpha}_1 + 2\vec{\alpha}_2}] &= -\frac{\sqrt{2}}{4} e_{\vec{\alpha}_1 + 3\vec{\alpha}_2} & [e_{-\vec{\alpha}_2}, e_{-\vec{\alpha}_1 - 2\vec{\alpha}_2}] &= -\frac{\sqrt{2}}{4} e_{-\vec{\alpha}_1 - 3\vec{\alpha}_2} \\
[e_{\vec{\alpha}_2}, e_{\vec{\alpha}_1 + 3\vec{\alpha}_2}] &= 0 & [e_{-\vec{\alpha}_2}, e_{-\vec{\alpha}_1 - 3\vec{\alpha}_2}] &= 0.
\end{aligned}$$

And we clearly get that if we switch the entries in the commutator that the coefficient will differ by a minus sign $[e_{\alpha}, e_{\beta}] = -[e_{\beta}, e_{\alpha}]$.

Part 6

Some of the vanishing Weyl commutators are

$$\begin{aligned}
[e_{\alpha_1}, e_{\alpha_1 + \alpha_2}] &= 0 \\
[e_{\alpha_1, \alpha_1 + 2\alpha_2}] &= 0 \\
[e_{\alpha_2}, e_{\alpha_1 + 3\alpha_2}] &= 0 \\
[e_{\alpha_1 + 3\alpha_2}, e_{2\alpha_1 + 3\alpha_2}] &= 0 \\
[e_{2\alpha_1 + 3\alpha_2}, e_{\alpha_1 + 2\alpha_1}] &= 0.
\end{aligned}$$

All of these vanish due to the fact that e_Λ (where Λ is the sum of the subscripts in the commutators i.e roots) form a vector which isn't a root and thus it must vanish.

Part 7

We know that

$$[E_{\vec{\alpha}}, E_{\vec{\beta}}] = \begin{cases} E_{\vec{\alpha}+\vec{\beta}} & \text{if } (\vec{\alpha} + \vec{\beta}) \text{ is a root} \\ 0 & \text{otherwise} \end{cases} \quad (27)$$

Thus we have closure under a lie bracket so thus it is a lie subalgebra.

First we will show that the lie subalgebra is nilpotent as that would imply that it is solvable

For a positive root we can write the root decomposed into some positive coefficients multiplied by the simple roots i.e.

$$\vec{\alpha} = \sum_i m_i \alpha_i \quad (28)$$

where m_i are the coefficients and α_i are the simple roots here $i = 2$.

We can call the sum of these coefficients H

$$H = \sum_i m_i \quad (29)$$

Since by definition these m_i are positive thus $H \geq 0$.

When we take the commutator of two positive weyl elements $E_{\vec{\alpha}}$ and $E_{\vec{\beta}}$ we get $cE_{\vec{\alpha}+\vec{\beta}}$ where c is a constant. This gives a monotonically increasing sequence, but our algebra is finite. Thus at some n our sequence of commutators will vanish. Therefore we have a nilpotent subalgebra which also implies that it is solvable.

Part 8

If we extend the set to include the cartan subalgebra then we have:

$$[H_i, H_j] = 0.$$

which is in the cartan set by definition.

$$[H_i, E_\alpha] = \alpha(H_i) E_\alpha.$$

Which is closed

$$\left[E_{\vec{\alpha}}, E_{\vec{\beta}} \right] = \begin{cases} E_{\vec{\alpha}+\vec{\beta}} & \text{if } (\vec{\alpha} + \vec{\beta}) \text{ is a root} \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

thus the positive weyl elements together with the cartan subalgebra is closed and obeys the lie bracket so it is a lie subalgebra which I will denot by $\mathcal{L}$.

Now if we look at if the lie subalgebra is sovlable and/or nilpotent.

By definition the cartan-cartan commutator vanishes and we showed above that for two postiive weyl elements $\exists k \in \mathbb{N}$ such that it vanishes. Since no matter how many times we compute the commutator between a pos weyl element and a cartan element it wil never vanish (it will just modify the constant $\alpha(H_i)$ which is always non-zero) thus it is not nilpotent.

If we look at the cartan subalgebra $[H_i, H_j] = 0$ we see by definition the commutator vanishes. Thus $\exists$ some $\lambda_i, \lambda_j \in \mathcal{L}$ such that $[\lambda_i, \lambda_j] = 0$. Thus we have that the subalgebra is solvable.